CREDIT RATING DYNAMICS AND DEFAULT RISK

A Markov Chain Simulation Study

Research Portfolio Report

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Companion workbook: 10_Credit_Rating_Simulation_Portfolio.xlsx

Abstract

This study investigates **corporate credit-rating migration** and **default risk** using **discrete-time Markov chains** and **Monte Carlo simulation**. A **three-state model** distinguishes **A**-rated firms, **B**-rated firms, and **Default**. Under an **absorbing**-default specification, rating deterioration accumulates across a fixed portfolio; under a **replacement** specification, defaulted positions are replenished with newly rated firms.

The companion Excel workbook simulates 1,000 rating paths over 200 annual states and compares conditional migration probabilities, portfolio composition, long-run rating shares, and first-passage times against analytical benchmarks. In the saved workbook snapshot, the **absorbing portfolio** reaches **99.8% Default** at year 200, whereas the **replacement model** yields **50.6% A, 46.3% B,** and **3.1% Default**, close to its stationary distribution. For the subset initially rated A, the workbook reports a simulated mean of **28.434** periods before first Default, compared with an infinite-horizon analytical expectation of 31.298 transitions.

These results illustrate how model assumptions fundamentally alter the economic interpretation of long-run credit risk.

Keywords: credit rating migration; Markov chains; Monte Carlo simulation; absorbing states; stationary distribution; first-passage time.

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1. Introduction

Credit ratings summarize the assessed ability of an issuer to meet its financial obligations, but credit quality is not static. An issuer may retain its rating, improve or deteriorate, or default. **Credit migration** therefore matters not only for assessing the risk of an individual issuer but also for forecasting the changing risk profile of a portfolio. A **transition matrix** provides a parsimonious description of these movements: conditional on the current rating, it assigns probabilities to each possible rating in the following year.

The **central modeling question is whether a portfolio should be treated as a fixed cohort of firms or as a continuing economic sector in which failed firms are replaced**. A **fixed** cohort with absorbing Default must eventually disappear into Default, even though actual sectors continue to contain operating firms. A **replacement** mechanism resolves this apparent tension by separating the fate of an individual firm from the evolving composition of rated positions. The analysis develops both representations, compares their forecasts, and examines how the expected time to a first default depends on initial credit quality.

The transition probabilities in this study are stipulated scenario parameters, not estimates obtained from historical rating-agency records. The objective is methodological: to demonstrate reproducible simulation logic, interpret state distributions correctly, and assess agreement between simulated and analytical outcomes. Prior credit-risk research likewise recognizes both the usefulness and restrictiveness of time-homogeneous Markov specifications (Kiefer and Larson, 2007; Lencastre et al., 2014).

2. Research Questions and Analytical Framework

The study addresses **six linked questions**: **(a) conditional migration from initially A- or B-rated firms; (b) portfolio composition at specified future years; (c) the consequences of absorbing Default; (d) a replacement transition mechanism; (e) its year-200 and stationary rating distribution**, including the default-event share; and **(f) the expected number of transitions before an initially A-rated firm first defaults**.

2.1. Rating states and transition probabilities

Let $X_t \in \{1, 2, 3\}$ denote a firm's rating in year t, where 1 = A, 2 = B, and 3 = Default. For the original model, the **one-year transition matrix** is:

Current state	Next: A	Next: B	Next: Default
A	0.81	0.18	0.01
B	0.17	0.77	0.06
Default	0.00	0.00	1.00

Each row sums to one. The third row makes Default absorbing: once a firm defaults, it remains in state 3 in the original model. The Markov assumption makes the next-year rating conditional on the current rating rather than the full rating history. Time homogeneity holds the transition probabilities fixed over the entire horizon.

The initial portfolio distribution is $f_1 = (0.30, 0.70, 0)$, corresponding to a 30% A and 70% B mix among 1,000 simulated firms. With X_1 defined as the initial state, the analytical distribution in year t is $f_t = f_1$

P^{t-1}. Hence year 2 represents one transition after the initial rating, and year 5 represents four transitions. Conditional forecasts from a purely A- or B-rated initial cohort follow the same indexing.

2.2. Firm replacement and a stable rated population

To model continuing participation in a sector, a position in Default is replaced in the following year by a new A-rated or B-rated firm, each with probability 0.5. This yields the **replacement transition matrix**:

Current position	Next: A	Next: B	Next: Default
A	0.81	0.18	0.01
B	0.17	0.77	0.06
Default	0.50	0.50	0.00

This is a model of rated positions, not a claim that a defaulted company recovers immediately. A Default state represents the period in which a default event is recorded; replacement takes place in the subsequent transition. Consequently, the stationary Default-state share is interpretable as the long-run fraction of positions registering a default event in a year under the model's timing convention.

3. Simulation Design and Excel Implementation

3.1. Sampling procedure

The workbook implements N = 1,000 simulated paths over T = 200 annual states. It converts each probability row to a cumulative distribution with a leading zero to support Excel's MATCH lookup. The first state is drawn from the initial cumulative distribution, and subsequent states are drawn conditionally from the cumulative row corresponding to the previous state. In conceptual Excel notation:

Initial rating: MATCH(RAND(), F1_CDF)

Subsequent rating: MATCH(RAND(), INDEX(Q, previous_rating, 0))

The original and replacement specifications use separate simulation paths and their respective cumulative transition matrices. For each year, the empirical share of rating k is the number of simulated paths in state k divided by 1,000. Conditional Question (a) estimates divide by the number of paths with the specified initial rating rather than by all 1,000 paths.

3.2. Validation and analytical comparison

Analytical results are calculated by multiplying the appropriate transition matrix, while the stationary distribution π of the replacement model satisfies $\pi P_{new} = \pi$ and $\pi_A + \pi_B + \pi_{\text{Default}} = 1$. These are distinct from the workbook's random-draw estimates. Absolute simulation-to-theory differences are descriptive checks, not evidence that the input transition matrix was empirically estimated.

All empirical figures and reported simulation values below are drawn from the saved calculation results in the final companion workbook. The charts are report-format reproductions of those saved state counts and do not constitute additional simulation runs. Because the workbook uses volatile RAND() formulas, reopening or recalculating it may generate a different valid realization.

4. Results

4.1. Question (a): conditional credit-rating migration

Initial rating	Year	Simulated A / B / Default	Analytical A / B / Default
A	2	84.38%/14.93%/0.69%	81.00%/18.00%/1.00%
A	5	58.33%/33.33%/8.33%	54.79%/37.26%/7.94%
B	2	15.03%/78.65%/6.32%	17.00%/77.00%/6.00%
B	5	33.85%/48.03%/18.12%	35.19%/46.51%/18.29%

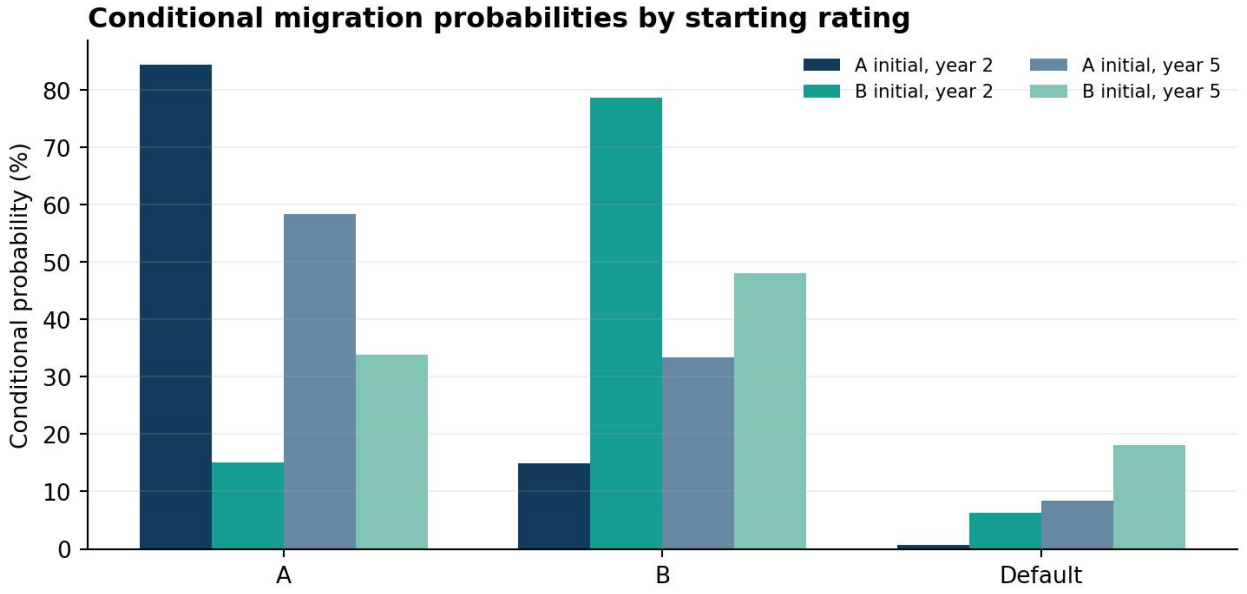


Figure 1. Conditional credit migration in years 2 and 5, grouped by initial rating. Source: saved Excel sample paths.

An **initially B-rated** firm has an analytical 6.00% probability of Default in **year 2** and 18.29% in **year 5**. An **initially A-rated** firm has corresponding analytical Default probabilities of 1.00% and 7.94%. The workbook's conditional simulation estimates are reported in Table 3 and Figure 1. The larger longer-horizon probabilities reflect repeated exposure to transitions into Default; differences between simulation and theory reflect finite sample counts within the A- and B-initial subgroups.

4.2. Question (b): evolution of the fixed portfolio

Year	Simulated A / B / Default	Analytical A / B / Default
2	37.30%/58.30%/4.40%	36.20%/59.30%/4.50%
5	42.20%/42.40%/15.40%	41.07%/43.74%/15.19%
10	38.10%/33.30%/28.60%	36.76%/34.27%/28.97%
50	9.70%/9.20%/81.10%	9.30%/8.54%/82.16%
200	0.10%/0.10%/99.80%	0.05%/0.05%/99.90%

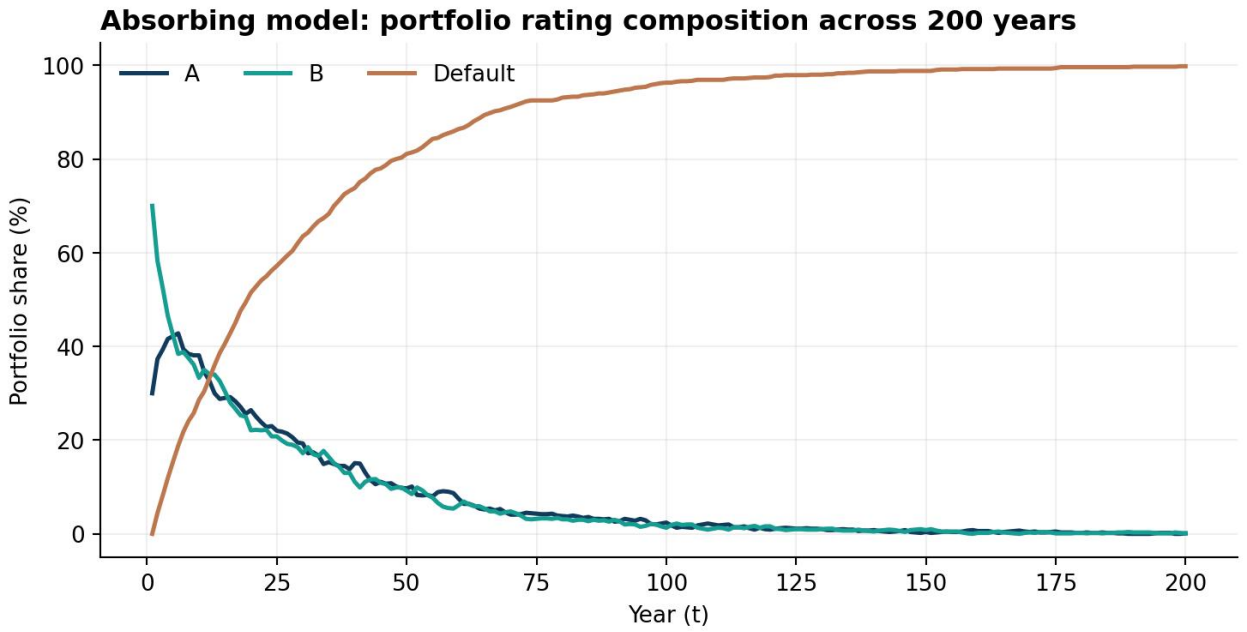


Figure 2. Year-by-year rating composition of the absorbing-default portfolio, derived from the workbook's saved portfolio paths.

The original portfolio is initially concentrated in A and B, but **Default becomes progressively more prevalent**. In the saved realization, the year-200 distribution is 0.1% A, 0.1% B, and **99.8% Default**, compared with an analytical Default share of approximately 99.90%. Figure 2 uses all 200 saved annual states to make the cumulative effect visible; the five specified forecast horizons appear in Table 4.

4.3. Question (c): why the absorbing cohort disappears

The long-run outcome follows directly from the model structure. Both nondefault states can reach Default with positive probability, while Default has no exit. The only stationary distribution for the absorbing model is therefore (0, 0, 1). This result concerns a fixed set of tracked firms, not the disappearance of all firms from the real economy. Entry, exit, replacement, and changes in the composition of the rated universe require a different population-level mechanism.

4.4. Questions (d) and (e): replacement and long-run composition

Rating	Simulated, year 200	Analytical stationary	Absolute difference
A	50.60%	50.24%	0.36 pp
B	46.30%	46.47%	0.17 pp
Default	3.10%	3.29%	0.19 pp

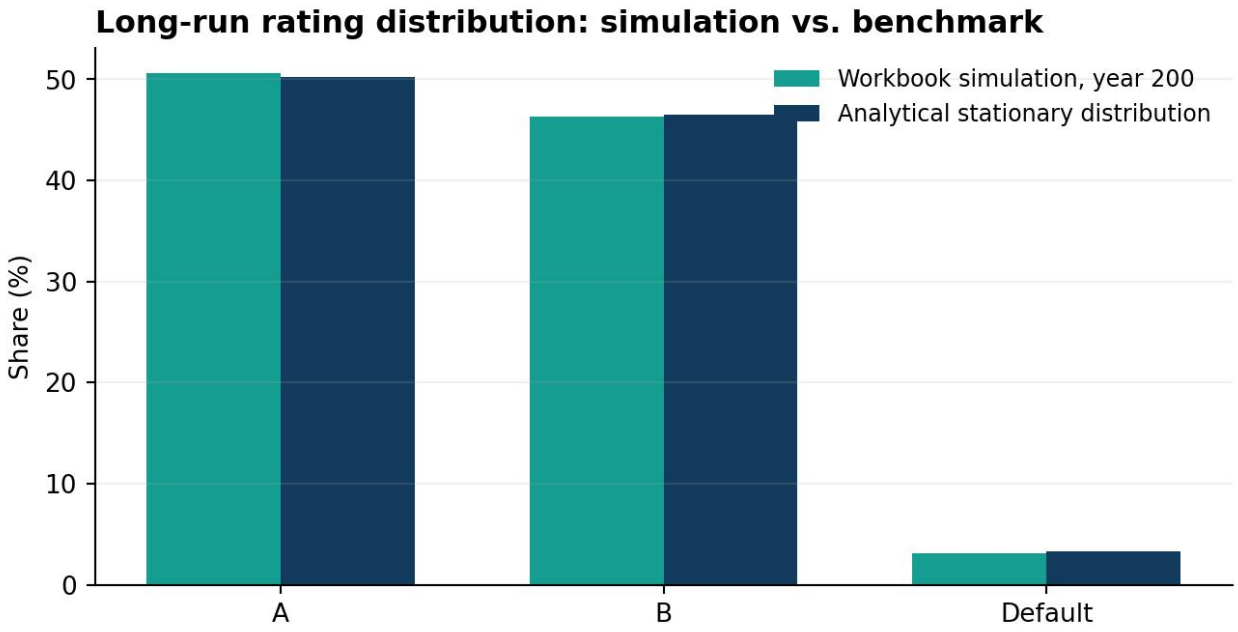


Figure 3. Replacement-model year-200 shares versus the analytical stationary distribution.

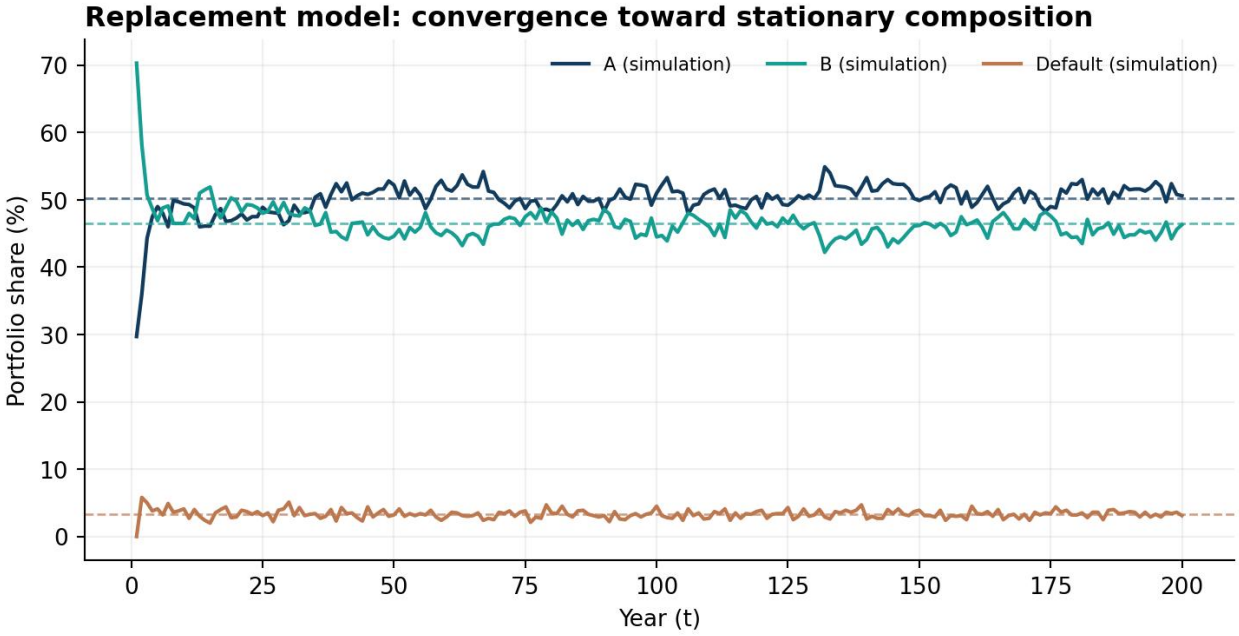


Figure 4. Convergence of replacement-model rating shares over all 200 saved annual states. Dashed lines mark analytical stationary shares.

The replacement-model snapshot has **50.6% A, 46.3% B,** and **3.1% Default** at **year 200**. The stationary benchmark is approximately 50.24% A, 46.47% B, and 3.29% Default. Deviations are 0.36, 0.17, and 0.19 percentage points, respectively. These discrepancies are small relative to the typical sampling fluctuation associated with 1,000 independently simulated positions at a fixed date.

Figure 4 distinguishes convergence of the distribution from stability of individual ratings. Firms continue to migrate and default, while the cross-sectional shares fluctuate around stable long-run levels. The stationary A and B shares are close but not equal, so their paths need not overlap. Under the stated timing convention, the approximately 3.29% theoretical Default-state share represents the long-run annual incidence of

recorded defaults among rated positions; the 3.1% simulated year-200 share is a single-year estimate rather than a historical annual average.

4.5. Question (f): time to first Default from an A rating

The first-passage statistic isolates paths initially in state A and records the first position at which Default appears. The workbook then subtracts one to express the result as the number of transitions elapsed after the initial state. For a path with no Default during the 200-state window, the **first-default position** is capped at 200 and the recorded **elapsed count** is 199. This is a finite-horizon, capped statistic rather than an uncensored lifetime measure.

Measure	Workbook / analytical result
A-initial paths in saved workbook	297
Simulated mean elapsed transitions before first Default	**28.434**
Infinite-horizon analytical mean from A	31.298
Difference (simulation − analytical)	−2.863

The theoretical mean follows from first-step equations: $m_A = 1 + 0.81m_A + 0.18m_B$ and $m_B = 1 + 0.17m_A + 0.77m_B$, with zero remaining time after Default. Solving gives $m_A \approx 31.298$ transitions. The saved conditional estimate of 28.434 is lower; the gap is consistent with a finite set of A-initial paths and possible horizon truncation, and should not be presented as a statistically significant difference without a dedicated uncertainty analysis. The replacement process does not change the time to first Default before that first event, although subsequent transitions represent replacement firms rather than the original obligor.

5. Interpretation and Managerial Implications

5.1. Credit monitoring is horizon-dependent

One-year transition probabilities are insufficient to describe **multiyear** portfolio exposure. Even an initially strong rating can experience migration into a weaker category before default. Risk managers can use **conditional multiyear distributions** to assess monitoring intensity, credit limits, and horizon-specific scenario exposure, provided that the transition probabilities are calibrated to a relevant portfolio.

5.2. Cohort risk and sector composition answer different questions

The **absorbing specification** is appropriate for following the ultimate fate of an original cohort. The **replacement specification** instead represents a continuing stock of rated positions. Its stable rating composition does not imply low lifetime default risk for a particular issuer. For strategic planning, this distinction separates questions about loss accumulation among existing obligors from questions about sustainable portfolio or sector composition after turnover.

5.3. Default-event flow and capital planning

In the replacement model, the stationary **Default share** offers a stylized estimate of the fraction of positions experiencing a default event in a period. Translating this fraction into expected credit loss, required reserves, or capital would require additional inputs—exposure at default, loss given default, dependence across

issuers, and possibly discounting—that are not included in the workbook. No monetary loss forecast is inferred from the reported shares.

5.4. Convergence and uncertainty

The replacement distribution approaches its stationary composition as the influence of initial ratings diminishes. The convergence plot is a dynamic validation of this behavior, not a formal proof of model adequacy for real credit markets. For orientation, a binomial normal-approximation half-width at 95% for an A share near 50.24% with N = 1,000 is about 3.10 percentage points. This is an illustrative cross-sectional sampling benchmark under independent-position assumptions, not a confidence interval for estimated transition probabilities or a guarantee of coverage for a correlated real-world portfolio.

6. Limitations and Directions for Further Research

First, the three-rating state space compresses meaningful differences within investment-grade, speculative-grade, and distressed issuers.

Second, transition probabilities are assumed constant across years, despite the possibility of business-cycle dependence, issuer heterogeneity, and rating momentum. Empirical research has questioned the reliability of homogeneous Markov assumptions across different credit markets and horizons (Kiefer and Larson, 2007; Lencastre et al., 2014).

Third, simulated paths are treated as independent, excluding common shocks that can produce correlated downgrades and defaults.

Fourth, the replacement mechanism preserves the number of modeled positions but does not explicitly model firm formation, financing constraints, or changes in the total outstanding amount of debt.

Fifth, the 200-state observation window and capped first-passage statistic can bias the conditional mean relative to an infinite-horizon expectation when defaults are not observed within the window.

Finally, results are conditional on a specified transition matrix rather than externally validated historical data. Extensions could estimate migration matrices from issuer-level histories, introduce macroeconomic regimes and correlated shocks, conduct sensitivity analysis, and report replicated simulation confidence intervals.

7. Conclusion

The project demonstrates that the interpretation of **long-run credit risk depends critically on the modeled population**. For a **fixed** cohort, **absorbing Default accumulates until nearly all firms are in Default**. With **replacement**, the same A- and B-rating transition probabilities **support a stable long-run composition of rated positions**. The workbook's year-200 replacement shares closely match the analytical stationary distribution, while conditional migration and first-passage calculations quantify risks faced by issuers with different starting ratings.

Together, the Excel model and its numerical validation provide a transparent foundation for **scenario analysis** and a clear illustration of the distinction between **individual default risk** and **population-level equilibrium**.

References

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Lencastre, P., Raischel, F., Lind, P. G., & Rogers, T. (2014). Are credit ratings time-homogeneous and Markov? arXiv:1403.8018. https://arxiv.org/abs/1403.8018

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Appendix A. Workbook traceability

Report element	**Companion workbook source**
Model parameters	1.Credit_Rating: C5:E18
Conditional migration (a)	1.Credit_Rating: HF2:HF7; HH2:HH7
Portfolio horizons (b)	03 | Results & Validation: B25:H29
Replacement distribution (e)	1.Credit_Rating: XT1:XT3
First-passage statistic (f)	1.Credit_Rating: PU2:PU1001; XM2:XN1001; XX1
Analytical benchmarks	03 | Results & Validation: F8:F19; F25:H29; D36:D38; C44
200-year trajectory figures	Saved sample-path states in 1.Credit_Rating (original and replacement simulations)

Reproducibility note. All simulated percentages, conditional counts, and trajectory plots in this report are based on the saved cell values of the specified final workbook. They may change if its RAND()-based formulas recalculate. The theoretical matrices, stationary probabilities, and analytical first-passage expectation do not depend on the random realization.